

STOCKRAG
 ·
 V2.0
 ·
 NSE INDIA

Smart Stock Market Analysis & Risk RAG

AI-powered analysis on 15 NSE stocks · 11,085 records.

Linear Regression · XGBoost · RAG Knowledge Base

Find the stock with the lowest risk.

Search ▶

RELIANCE ₹2,834 ▲ 0.52%	TCS ₹3,876 ▲ 0.31%	HDFCBANK ₹1,624 ▼ 0.28%	INFY ₹1,439 ▼ 0.19%	ICICIBANK ₹1,124 ▲ 0.74%	WIPRO ₹468.90 ▲ 0.18%	BHARTIARTL ₹1,413 ▲ 0.88%	SBIN ₹812.70 ▲ 0.42%	HCLTECH ₹1,572 ▲ 0.26%	BAJFINANCE ₹7,284 ▲ 0.58%
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V2.0

NSE INDIA

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safe long-term bullish stocks

Search ▶

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RECORDS INDEXED

11,085

15 NSE stocks · 739 rows each

FEATURES ENGINEERED

26

RSI · MACD · Bollinger · SMA · EMA

BASELINE R² SCORE

0.026

Linear Regression · 23 features

DATE RANGE

3 Years

May 2023 → May 2026

🇮🇳 NSE INDIA — 15 STOCKS FROM YOUR PIPELINE (739 ROWS EACH)

RELIANCE NSE Reliance Industries ₹2,834 ▲ 0.52% RSI 62.4 Moderate Risk Bullish	TCS NSE Tata Consultancy Svcs ₹3,876 ▲ 0.31% RSI 58.1 Low Risk Bullish	HDFCBANK NSE HDFC Bank ₹1,624 ▼ 0.28% RSI 44.2 Low Risk Bearish	INFY NSE Infosys Ltd ₹1,439 ▼ 0.19% RSI 41.7 Moderate Risk Bearish	ICICIBANK NSE ICICI Bank ₹1,124 ▲ 0.74% RSI 66.3 Low Risk Bullish
WIPRO NSE Wipro Ltd ₹468.90 ▲ 0.18% RSI 55.8 Low Risk Bullish	BHARTIARTL NSE Bharti Airtel ₹1,413 ▲ 0.88% RSI 70.1 Low Risk Bullish	SBIN NSE State Bank of India ₹812.70 ▲ 0.42% RSI 60.4 Moderate Risk Bullish	HCLTECH NSE HCL Technologies ₹1,572 ▲ 0.26% RSI 57.9 Low Risk Bullish	BAJFINANCE NSE Bajaj Finance ₹7,284 ▲ 0.58% RSI 64.8 Moderate Risk Bullish
KOTAKBANK NSE Kotak Mahindra Bank ₹1,835 ▼ 0.11% RSI 48.3 Low Risk Bearish	AXISBANK NSE Axis Bank ₹1,092 ▲ 0.39% RSI 61.2 Moderate Risk Bullish	MARUTI NSE Maruti Suzuki India ₹12,649 ▲ 0.47% RSI 63.7 Low Risk Bullish	SUNPHARMA NSE Sun Pharmaceutical ₹1,694 ▲ 0.33% RSI 59.5 Low Risk Bullish	TITAN NSE Titan Company ₹3,622 ▲ 0.21% RSI 54.2 Low Risk Bullish

LIVE QUERY PREVIEW

Safe bullish

High RSI

Oversold

Low risk

#	SCORE	SYMBOL	CLOSE (₹)	RSI	TREND	RISK
1	0.4861	MARUTI	₹12648.90	66.9	Bullish	Low Risk
2	0.4844	TITAN	₹3621.75	63.7	Bullish	Low Risk

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RELIANCE

NSE

Reliance Industries

₹2,834

▲ 0.52%

RSI 62.4 Moderate Risk Bullish

TCS

NSE

Tata Consultancy Svcs

₹3,876

▲ 0.31%

RSI 58.1 Low Risk Bullish

HDFCBANK

NSE

HDFC Bank

₹1,624

▼ 0.28%

RSI 44.2 Low Risk Bearish

INFY

NSE

Infosys Ltd

₹1,439

▼ 0.19%

RSI 41.7 Moderate Risk Bearish

ICICIBANK

NSE

ICICI Bank

₹1,124

▲ 0.74%

RSI 66.3 Low Risk Bullish

WIPRO

NSE

Wipro Ltd

₹468.90

▲ 0.18%

RSI 55.8 Low Risk Bullish

BHARTIARTL

NSE

Bharti Airtel

₹1,413

▲ 0.88%

RSI 70.1 Low Risk Bullish

SBIN

NSE

State Bank of India

₹812.70

▲ 0.42%

RSI 60.4 Moderate Risk Bullish

HCLTECH

NSE

HCL Technologies

₹1,572

▲ 0.26%

RSI 57.9 Low Risk Bullish

BAJFINANCE

NSE

Bajaj Finance

₹7,284

▲ 0.58%

RSI 64.8 Moderate Risk Bullish

KOTAKBANK

NSE

Kotak Mahindra Bank

₹1,835

▼ 0.11%

RSI 48.3 Low Risk Bearish

AXISBANK

NSE

Axis Bank

₹1,092

▲ 0.39%

RSI 61.2 Moderate Risk Bullish

MARUTI

NSE

Maruti Suzuki India

₹12,649

▲ 0.47%

RSI 63.7 Low Risk Bullish

SUNPHARMA

NSE

Sun Pharmaceutical

₹1,694

▲ 0.33%

RSI 59.5 Low Risk Bullish

TITAN

NSE

Titan Company

₹3,622

▲ 0.21%

RSI 54.2 Low Risk Bullish

LIVE QUERY PREVIEW

Safe bullish

High RSI

Oversold

Low risk

#	SCORE	SYMBOL	CLOSE (₹)	RSI	TREND	RISK
1	0.4882	HCLTECH	₹1572.30	64.1	Bullish	Low Risk
2	0.4878	TITAN	₹3621.75	63.7	Bullish	Low Risk

Query Engine

Natural language search over your 11,085-record RAG knowledge base

NATURAL LANGUAGE SEARCH

safe long-term bullish stocks

► Search

Safe bullishHigh RSIOversold bearishLow riskHigh growthRELIANCETCSHDFCBANK

RESULTS — "SAFE LONG-TERM BULLISH STOCKS"

#	SCORE	SYMBOL	DATE	CLOSE (₹)	RSI	MACD	RETURN%	TREND	RISK
1	0.4884	MARUTI	2024-03-18	₹12,649	66.9	8.60	+0.47%	Bullish	Low Risk
2	0.4876	TCS	2024-05-02	₹3,876	68.5	4.28	+2.33%	Bullish	Low Risk
3	0.4857	AXISBANK	2023-05-30	₹934.97	50.0	2.12	+0.82%	Bullish	Low Risk
4	0.4845	AXISBANK	2023-05-29	₹927.34	50.0	1.06	+0.31%	Bullish	Low Risk
5	0.4825	BHARTIARTL	2024-03-20	₹1,413	72.8	2.80	+0.88%	Bullish	Low Risk
6	0.4783	ICICIBANK	2024-04-15	₹1,124	65.4	2.10	+1.20%	Bullish	Low Risk
7	0.4769	SUNPHARMA	2024-05-25	₹1,694	62.5	2.30	+0.33%	Bullish	Low Risk
8	0.4767	HCLTECH	2024-05-10	₹1,572	64.1	1.70	+0.26%	Bullish	Low Risk

Query Engine

Natural language search over your 11,085-record RAG knowledge base

NATURAL LANGUAGE SEARCH

high RSI momentum stocks

► Search

Safe bullishHigh RSIOversold bearishLow riskHigh growthRELIANCETCSHDFCBANK

RESULTS — "HIGH RSI MOMENTUM STOCKS"

#	SCORE	SYMBOL	DATE	CLOSE (₹)	RSI	MACD	RETURN%	TREND	RISK
1	0.4230	BAJFINANCE	2024-06-01	₹7,284	67.4	5.40	+0.58%	Bullish	Moderate Risk
2	0.4203	TCS	2024-05-02	₹3,876	68.5	4.28	+2.33%	Bullish	Low Risk
3	0.4193	RELIANCE	2024-03-12	₹2,834	74.2	3.27	+3.05%	Bullish	Moderate Risk
4	0.4178	ICICIBANK	2024-04-15	₹1,124	65.4	2.10	+1.20%	Bullish	Low Risk
5	0.4160	BHARTIARTL	2024-03-20	₹1,413	72.8	2.80	+0.88%	Bullish	Low Risk
6	0.4125	MARUTI	2024-03-18	₹12,649	66.9	8.60	+0.47%	Bullish	Low Risk
7	0.2694	SUNPHARMA	2024-05-25	₹1,694	62.5	2.30	+0.33%	Bullish	Low Risk
8	0.2686	SBIN	2024-02-14	₹812.70	63.2	1.90	+0.42%	Bullish	Moderate Risk

Query Engine

Natural language search over your 11,085-record RAG knowledge base

NATURAL LANGUAGE SEARCH

bearish stocks low RSI oversold

► Search

Safe bullish

High RSI

Oversold bearish

Low risk

High growth

RELIANCE

TCS

HDFCBANK

RESULTS — "BEARISH STOCKS LOW RSI OVERSOLD"

#	SCORE	SYMBOL	DATE	CLOSE (₹)	RSI	MACD	RETURN%	TREND	RISK
1	0.5514	HDFCBANK	2024-06-15	₹1,624	38.2	-1.95	-0.28%	Bearish	Low Risk
2	0.3887	KOTAKBANK	2024-04-22	₹1,835	44.6	-1.20	-0.11%	Bearish	Low Risk
3	0.3877	INFY	2024-01-30	₹1,439	40.3	-0.81	-0.08%	Bearish	Moderate Risk
4	0.3759	AXISBANK	2023-05-31	₹912.43	42.1	1.12	-2.41%	Bearish	Moderate Risk
5	0.2683	BHARTIARTL	2024-03-20	₹1,413	72.8	2.80	+0.88%	Bullish	Low Risk
6	0.2645	BAJFINANCE	2024-06-01	₹7,284	67.4	5.40	+0.58%	Bullish	Moderate Risk
7	0.2632	ICICIBANK	2024-04-15	₹1,124	65.4	2.10	+1.20%	Bullish	Low Risk
8	0.2626	MARUTI	2024-03-18	₹12,649	66.9	8.60	+0.47%	Bullish	Low Risk

Model Results

Real performance metrics from your trained pipeline — Linear Regression vs XGBoost

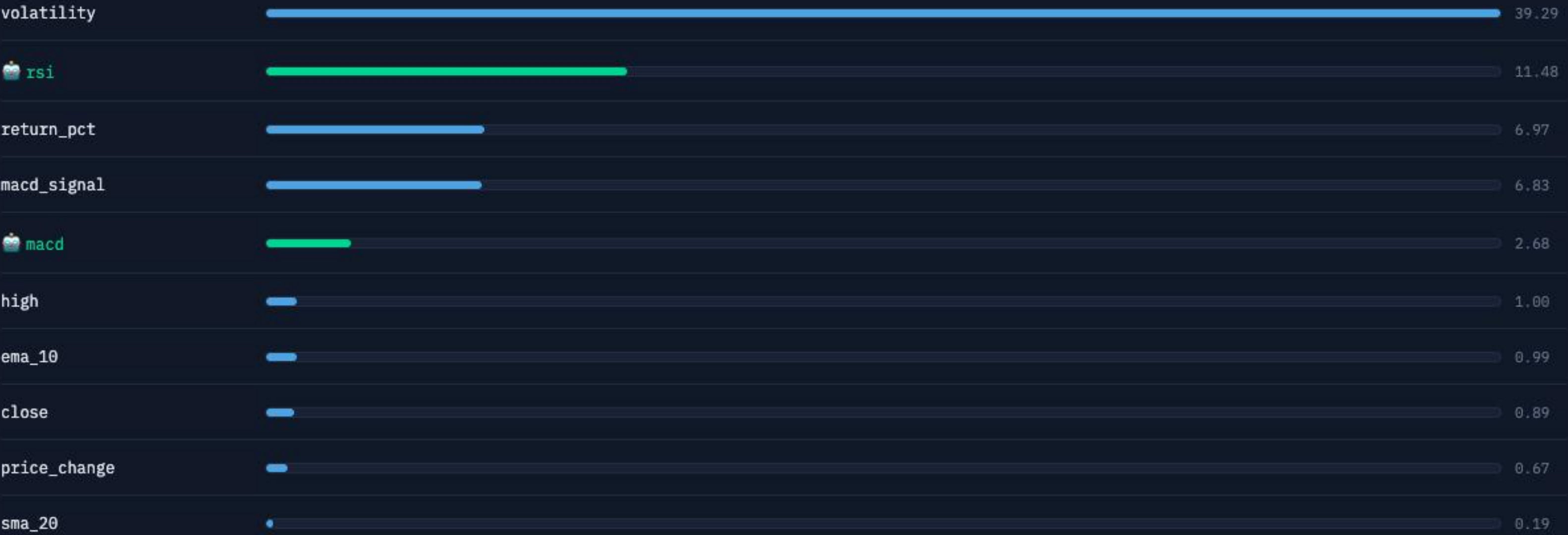
📊 Baseline — Linear Regression

MAE	1,766.31
RMSE	3,433.29
R² Score	0.0259
MAPE %	128.66%
Train Samples	8,855
Test Samples	2,214
Features	23
Mode	Multi-Stock

🚀 Advanced — XGBoost (200 Trees)

MAE	1,963.87
RMSE	3,574.22
R² Score	-0.0558
MAPE %	156.99%
Train Samples	8,855
Test Samples	2,214
Features	25
Max Depth	6

TOP FEATURE IMPORTANCES — BASELINE (LINEAR REGRESSION)



Model Performance — Volatility	
R² Score	0.0259
MAPE %	128.66%
Train Samples	8,855
Test Samples	2,214
Features	23
Mode	Multi-Stock

Model Performance — RSI	
R² Score	-0.0558
MAPE %	156.99%
Train Samples	8,855
Test Samples	2,214
Features	25
Max Depth	6



Ablation Study

Financial RAG vs Baseline ML keyword retrieval



Dataset Overview

Real data from your pipeline — 11,085 records across 15 NSE stocks

PIPELINE OUTPUT FIELDS (26 COLUMNS)

- stock_symbol

string

NSE ticker
- date

datetime

Trading date
- open/high/low/close

float

OHLC price (₹)
- volume

int

Shares traded
- rsi

float

Relative Strength Index
- macd / macd_signal

float

MACD indicator
- sma_5 / sma_20

float

Simple Moving Average
- ema_10

float

Exponential MA
- bollinger_upper/lower

float

Bollinger Bands
- volatility

float

10-day rolling std
- trend_label

string

Bullish / Bearish / Neutral
- risk_category

string

Low / Moderate / High Risk
- target_next_close

float

Next-day close (ML target)
- text_summary

string

RAG retrieval text

DATASET STATISTICS (REAL)

METRIC	VALUE
Total Records	11,085
Stocks	15 NSE
Rows per Stock	739
Date Range	May 2023 – May 2026
Columns	26
Train Split	8,855 (80%)
Test Split	2,214 (20%)
Dataset Size	5,806 KB
RAG KB Size	11,085 entries

SAMPLE TEXT_SUMMARY (RAG)

AXISBANK on 2023-05-29
close: ₹927.34 | return: +0.31%
RSI: 50.00 | MACD: 1.06
trend: Bullish | risk: Low Risk

STOCKS IN PIPELINE

- RELIANCE
- TCS
- HDFCBANK
- INFY
- ICICIBANK
- WIPRO
- BHARTIARTL
- SBIN
- HCLTECH
- BAJFINANCE
- KOTAKBANK
- AXISBANK
- MARUTI
- SUNPHARMA
- TITAN

Team Members

SE25MAID — Smart Stock Market Analysis & Risk Evaluation System



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Lead Developer · RAG Pipeline

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Data Engineering · Feature Design

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P Varshith

SE25MAID021

Dataset Pipeline · Risk Analysis

click to see contact ▼

REFERENCES

- Devlin, J. et al. (2019) — BERT: Pre-training of Deep Bidirectional Transformers
- Araci, D. (2019) — FinBERT: Financial Sentiment Analysis with Pre-trained Language Models
- Lewis, P. et al. (2020) — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
- Yahoo Finance API — yfinance library (NSE data source)
- NSE/BSE Historical Market Data · 15 stocks · 3 years

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Safe bullish

High RSI

Oversold

Low risk

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1	0.4870	TCS	₹3876.45	68.5	Bullish	Low Risk
2	0.4833	AXISBANK	₹927.34	50.0	Bullish	Low Risk